

Moustafa Mahmoud Ghoneim

Senior Software Engineer

✉ progmmgghoneim@gmail.com ☎ +201024459684 📍 Cairo ,Egypt 🌐 MustafaGhoneim

Personal Summary

- Results-driven Senior Software Engineer with over seven years of experience building scalable, maintainable back-end systems using Python, node js, and modern development practices. Specialized in designing micro services architectures, developing RESTful APIs, and integrating with third-party services. Proven ability to optimize performance, ensure system reliability, and drive business outcomes. Experienced in delivering high-impact software solutions across fintech, real estate, and SaaS sectors. Passionate about clean code, system design, and continuous learning in dynamic development environments.

Work Experience

Senior Software engineer - SAK company

Jan 2023 - Now

SAK Real Estate is a platform focused on marketing and selling real estate in Saudi Arabia, combining technical and financial solutions to deliver an end-to-end digital experience for sellers and buyers.

Tech stack: Python (FastAPI · Flask · Django) · API Style (Rest ·gRPC. GraphQL) · DB (Postgres. mongo DB). CRM (starpi JS). API Gateway (GO)

- **Architected scalable backend services** for payments, installments, and transaction management, designed for growth and clear service boundaries.
- **Implemented a centralized caching layer** shared by services to cut latency, reduce database load, and keep data access patterns consistent platform-wide.
- Designed and implemented an **API gateway in Go** to route, secure, and standardize traffic across backend services.
- Built **extensible integrations** for AI-driven capabilities and automation pipelines, keeping contracts and workflows maintainable as capabilities expand.
- **Improved performance and reliability** for high-throughput financial flows through profiling, caching, and resilient service design.
- **Contributed to system design** for AI-assisted decisioning and intelligent workflows, aligned with compliance and audit expectations.
- Automated core business processes using Camunda, improving traceability, handoffs, and operational consistency across teams.
- **Delivered a management system** for developer and vendor transactions, streamlining onboarding, tracking, and reporting across stakeholders.

Software engineer at paymob company

Nov 2019- Aug 2022

Tech stack: Python · Django · Django REST Framework · GraphQL · PostgreSQL

- Designed and integrated installment **payment methods** (valU, Souhoola, Contact, Forsa, Shahry), including provider contracts, reconciliation considerations, and error handling for production traffic.
- Implemented **cash-out flows** so customers could complete payments using reference numbers delivered via SMS, improving accessibility for non-digital-native users.
- Supported payment infrastructure for digital goods sold online and in person, including **POS** acceptance paths and operational monitoring.
- Contributed to **Uber payment integration** (card, wallet, and aggregator scenarios), collaborating on API design, edge cases, and rollout readiness.
- Built a **terminal management system** to govern POS transactions and branch-level terminal data, with reporting and export for operational and finance teams.
- Owned end-to-end PayPal integration: onboarding, payments, reporting, risk controls, and fee handling, aligned with compliance and reconciliation needs.
- **Migrated card authentication** from 3DS to 3DS2, reducing checkout friction while supporting richer authentication data for issuer decisioning.

Work Experience

Python Developer at M06 company

Sep 2019 - Jan 2020

- **Tech stack:** Python · Pandas · NumPy · Matplotlib · Selenium · BeautifulSoup
- Built tooling to retrieve and visualize historical website snapshots, enabling side-by-side comparison of pages across time periods for analysis and reporting.
- Developed browser automation (**Selenium**) to handle authenticated sessions and repeatable navigation flows for data extraction pipelines, with emphasis on stability, selectors, and error handling.
- Scraped and analyzed **cryptocurrency** market data, producing exploratory visualizations and summary statistics to support decision-making.
- Implemented a simple forecasting baseline for **Bitcoin price** movement using cleaned time-series features (modeling scope aligned to available data and validation constraints).

Education

Bachelor Degree in computer and information

Sep 2014 - Jun 2018

Faculty of Computer and Information , Computer Science Department,
Menofia University.

Generative AI Diploma

Comprehensive diploma program covering generative AI technologies,
machine learning fundamentals, and practical applications of AI in software
development.

Tech Skills

Programming languages

- Python
- JavaScript
- Go

Web Framework

- Fast API
- Django
- Flask
- Strapi

Database Management System

- MySQL
- Postgres
- Mongo DB
- SQL lite
- Redis

Architecture Styles

- Modular Monolith
- Microservices Architecture

Basic Knowledge

- Data Structures and Algorithms
- Problem Solving and Debugging
- Object Oriented Programming (OOP)
- Design Pattern
- SOLID
- Networking Basics

Languages

- Arabic : Fluent
- English: Intermediate